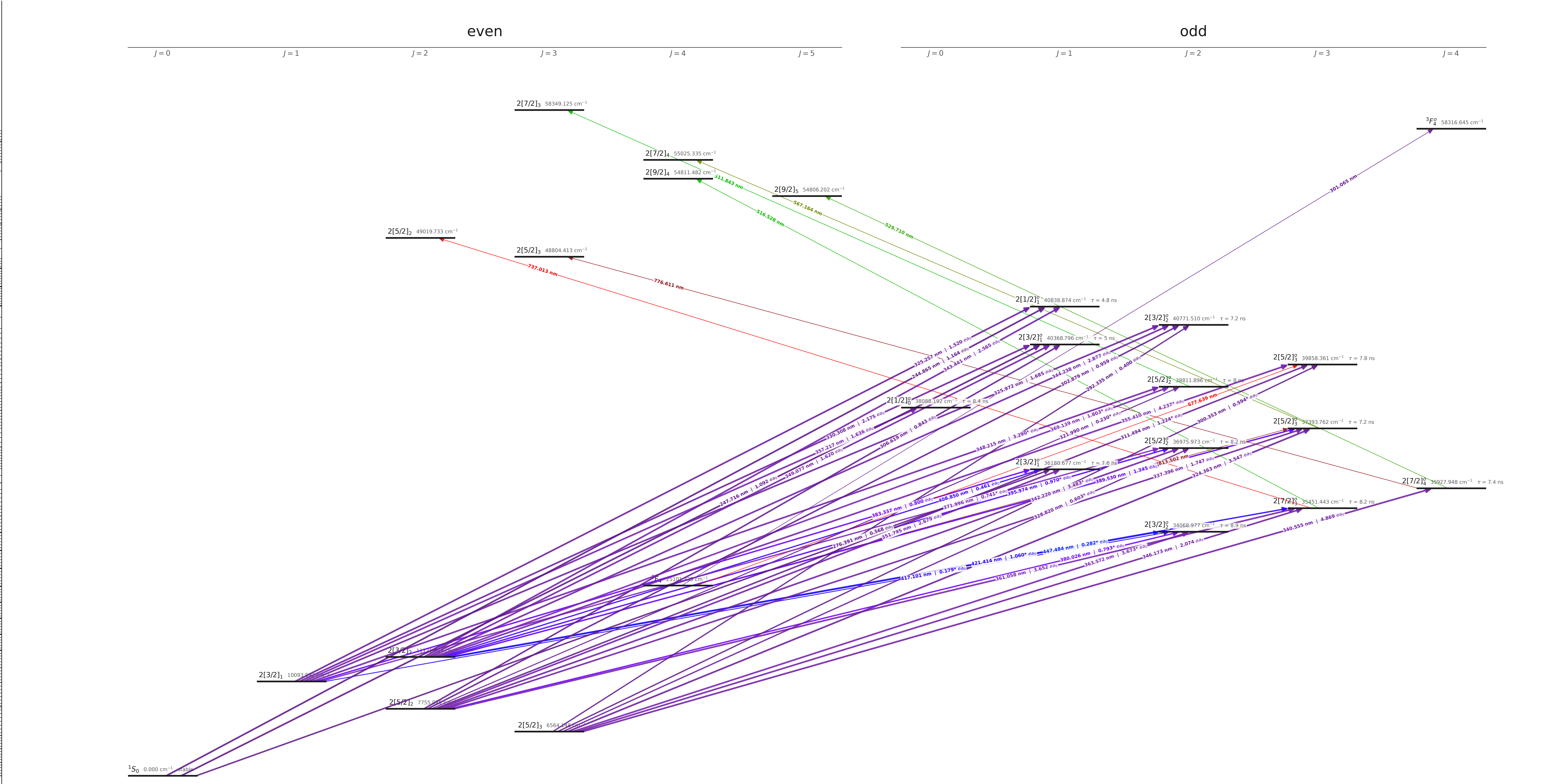


Pd (Pd I). Strongest classified transitions below 2 μm (NIST): 25 levels, 46 lines.

Pd⁺ ionisation limit 67241.14 cm⁻¹ = 8.33684 eV (148.718 nm)

Arrow label: vacuum wavelength reduced dipole matrix element $\langle J||er||J' \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U e U' \rangle$ (eao)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$1^1S_0 - 2[1/2]_2^1$	244.8647	244.7906	1224.3186	1.164	6.233e+07	33.16	28.6	ground-state resonance line 244.9 nm
$1^1S_0 - 2[3/2]_2^1$	247.7161	247.6413	1210.2261	1.092	5.300e+07	31.83	26.5	ground-state resonance line 247.7 nm
$1^1S_0 - 2[3/2]_2^3$	276.3906	276.3090	1084.6694	0.568	1.033e+07	20.94	7.9	ground-state resonance
$2[5/2]_3 - 2[7/2]_3^1$	304.5554	304.4577	880.3046	4.869	1.351e+08	21.51	100	closed J=3 $\rightarrow$ J=4 cycling transition from the metastable...
$2[3/2]_1 - 2[1/2]_1^1$	357.2169	357.1149	839.2450	1.636	1.190e+08	18.95	100	357.2 nm line used for collinear laser spectroscopy tests
$2[5/2]_3 - 2[7/2]_3^3$	361.0575	360.9546	830.3177	3.652	8.200e+07	19.41	67.2	361.1 nm
$2[5/2]_3 - 2[3/2]_2^1$	363.5725	363.4689	824.5740	3.673*	1.137e+08	17.88	100	363.5 nm line from metastable 3D3: main line for collinea...
$2[3/2]_1 - 2[5/2]_3^1$	369.1387	369.0337	812.1404	1.803*	2.618e+07	19.89	20.9	369.1 nm line used for collinear laser spectroscopy tests

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)							
215/2 ₁	-391.1790(10)	-652.906(15)	F=1/2: +3520.05	F=3/2: +3109.56	F=5/2: +3260.13	F=7/2: +1173.82	F=9/2: -586.49	F=11/2: -3097.07		
215/2 ₂	66.3590(10)	-398.192(10)	F=1/2: -743.25	F=3/2: -464.52	F=5/2: -99.53	F=7/2: +202.41	F=9/2: +232.25			
213/2 ₁	-83.5(3)	-304(3)	F=1/2: +371.63	F=3/2: +383.23	F=5/2: +326.52	F=7/2: +87.49	F=9/2: -493.52			
217/2 ₁	-115.3(3)	-497(4)	F=1/2: +854.80	F=3/2: +816.04	F=5/2: +701.74	F=7/2: +437.35	F=9/2: -81.50	F=11/2: -989.00		
217/2 ₂	-79.7(3)	-469.9(18)	F=3/2: +765.50	F=5/2: +704.70	F=7/2: +566.72	F=9/2: +298.69	F=11/2: -167.35	F=13/2: -914.48		
213/2 ₂	-126.2(6)	6.0(20)	F=3/2: +445.90	F=5/2: +121.40	F=7/2: -314.00					

Measured A, B constants; sources in the data file.

Transition	λ vac. (nm)	Isotopes	Shift (MHz)
$1^1S_0 - 2^1D_2$ $^{16}_2$	276.391	105-108	77(3)
$2^1S_0/2_1 - 2^1P^o_1$ $^{16}_2$	340.555	104-106	468.6(27)
$2^1S_0/2_1 - 2^1P^o_2$ $^{16}_2$	340.555	105-106	347(3)
$2^1S_0/2_1 - 2^1P^o_2$ $^{18}_2$	340.555	108-106	-505(4)
$2^1S_0/2_1 - 2^1P^o_2$ $^{14}_2$	340.555	110-106	-947(5)
$2^1S_0/2_1 - 2^1P^o_2$ $^{16}_2$	340.555	102-106	938(18)
$2^3P_1/2_1 - 2^1D_2$ $^{16}_2$	357.217	102-108	1446(4)
$2^3P_1/2_1 - 2^1D_2$ $^{18}_2$	357.217	104-108	958(3)
$2^3P_1/2_1 - 2^1D_2$ $^{14}_2$	357.217	105-108	842.0(20)
$2^3P_1/2_1 - 2^1D_2$ $^{16}_2$	357.217	106-108	490.0(20)
$2^3P_1/2_1 - 2^1D_2$ $^{16}_2$	357.217	110-108	-443(5)
$2^1S_0/2_1 - 2^1P^o_2$ $^{16}_2$	361.058	102-108	1413.6(14)
$2^1S_0/2_1 - 2^1P^o_2$ $^{18}_2$	361.058	104-108	948.4(11)
$2^1S_0/2_1 - 2^1P^o_2$ $^{16}_2$	361.058	105-108	844.7(9)
$2^1S_0/2_1 - 2^1P^o_2$ $^{16}_2$	361.058	106-108	486.2(11)
$2^1S_0/2_1 - 2^1P^o_2$ $^{16}_2$	361.058	110-108	-429.4(15)
$2^1S_0/2_1 - 2^3P_2$ $^{16}_2$	363.572	104-106	467.1(14)
$2^1S_0/2_1 - 2^3P_2$ $^{18}_2$	363.572	105-106	348.4(16)
$2^1S_0/2_1 - 2^3P_2$ $^{14}_2$	363.572	108-106	-497.2(15)
$2^1S_0/2_1 - 2^3P_2$ $^{16}_2$	363.572	110-106	-939.2(27)
$2^3P_1/2_1 - 2^1S_0$ $^{16}_2$	369.139	102-108	1337(4)
$2^3P_1/2_1 - 2^1S_0$ $^{18}_2$	369.139	104-108	889(4)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$